

Self-Attentive Sequential Recommendation Models Enriched with More Features

Trong Dang Huu Ho

School of Computer Science and Engineering,
International University, VNU-HCMC
Vietnam
Vietnam National University - HCMC
Vietnam
hdhtrong@hcmiu.edu.vn

Sang Thi Thanh Nguyen

School of Computer Science and Engineering,
International University, VNU-HCMC
Vietnam
Vietnam National University - HCMC
Vietnam
nttsang@hcmiu.edu.vn

Abstract

Recommender systems have become essential for alleviating information overload by providing personalized suggestions, with traditional collaborative filtering methods based on matrix factorization widely adopted but often struggling to capture dynamic user interests and complex item relationships. Recently, deep learning techniques utilizing the attention mechanism have shown promise in modeling sequential user behavior for recommendation tasks by leveraging self-attention to identify relevant historical interactions when predicting the next item of interest. However, these models primarily rely on raw interaction sequences, failing to exploit auxiliary information, such as user ratings and item categories that could further enhance recommendation accuracy. In this study, we propose a self-attentive sequential recommendation method that enriches input representations by incorporating user ratings to capture explicit preferences and item categories to capture semantic relationships between items, extending the self-attention architecture to attend to the combination of interaction sequences, ratings, and categories. Through extensive experiments on public datasets, we demonstrate that our proposed method exhibits relative strengths in capturing certain aspects of user-item relationships, leading to competitive performance compared to original models across different recommendation quality metrics.

CCS Concepts

• **Information systems** → Information retrieval; Retrieval models and ranking.

Keywords

Recommender Systems, Sequential Prediction, Self-attention, Self-Attentive Models

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1 Introduction

In the era of information overload, recommender systems have emerged as an indispensable tool for alleviating user challenges in navigating the vast online landscape [1]. The exponential growth of e-commerce platforms has created fertile ground for recommender system applications, providing personalized suggestions tailored to individual preferences while driving substantial business value [2]. Traditional recommendation techniques like collaborative filtering often struggle to capture dynamic user interests and complex item relationships, suffering from issues such as data sparsity, cold-start, scalability, and the inability to model long-range dependencies in user behaviors [3].

Deep learning advancements have transformed recommender systems by leveraging neural networks to identify intricate patterns in large datasets [4]. These advancements are particularly crucial for sequential recommendation, which aims to predict the next item a user will interact with based on their past behavior, enhancing user experience across domains like e-commerce, tourism, and entertainment [5]. Early sequential mining models using RNNs, and LSTMs captured sequential patterns but were limited by long-range dependencies and computational costs. The introduction of the Transformer architecture and self-attention mechanisms revolutionized the field, enabling effective long-range dependency modeling and parallelization [6]. Models such as SASRec [7] and TiSASRec [8] leverage self-attention to identify relevant historical interactions for next-item predictions. However, these models primarily utilize raw interaction sequences, neglecting auxiliary information like user ratings and item categories. More recent approaches, such as S3-Rec [9] and BERT4Rec [10], demonstrate the benefits of integrating additional contextual data.

This study proposes a novel self-attentive sequential recommendation method that enriches input representations and extends the self-attention architecture. Our model aims to capture intricate user-item relationships for more accurate recommendations. Through this study, we made some contributions:

- Introducing a novel self-attentive sequential recommendation method that incorporates auxiliary information like user ratings and item categories with competitive performance compared to the baselines.
- Addressing the limitations of existing self-attention mining models that rely solely on raw interaction sequences.

2 RELATED WORK

2.1 Traditional recommendation approaches

Traditional recommendation techniques like collaborative filtering (CF) and matrix factorization have been widely used in recommender systems [11, 12]. CF methods recommend items based on user preference similarities [13], but struggle with data sparsity, cold-start problems, and capturing dynamic interests. Matrix factorization models user-item interactions effectively but treats them as unordered, missing sequential and temporal patterns. Content-based filtering recommends items based on item features similar to those previously liked by the user [14] yet fails to capture complex user-item relationships. Markov Chains and Factorizing Personalized Markov Chains (FPMC) [15] model sequential patterns but cannot capture long-range dependencies. These traditional methods, while valuable, often fail to capture nuanced user preferences and tend to recommend primarily popular items, highlighting the need for further advancements in the field.

2.2 Sequential recommendations

Due to the limitations of traditional methods, sequential recommendation emerges as a distinct approach that considers the order in which items are consumed, allowing it to capture both short-term and long-term user preferences. The integration of deep learning techniques has significantly enhanced sequential recommendation models' ability to capture intricate patterns and dependencies within user-item interactions. Hidasi [16] introduced GRU4Rec, employing GRU-based RNNs for session-based recommendation. It focuses on capturing temporal dependencies within user sessions and predicts the next item based on recent interactions. NARM [17], proposed by Li et al., combines a hybrid encoder with an attention mechanism and GRU to model user sequential behavior. It creates a unified session representation and employs item-level attention to prioritize important items. STAMP [18], proposed by Liu et al., uses a multi-layer perceptron and attention network to separate short-term and long-term user interests within a session. It leverages the last interactive item for short-term interest and uses attention to extract long-term interest. The CASER model, introduced by Tang and Wang et al. [19], uses CNNs to embed recent item sequences into an "image"-like representation across temporal and latent spaces. It captures both general user preferences and sequential patterns with convolutional filters, focusing more on short-term preferences due to its local receptive fields. Addressing this, NextItNet [20] by Yuan et al. employs 1-D dilated convolutions to extend the model's reach, enhancing its ability to handle longer sequential contexts and recommend users with extensive interaction histories.

2.3 Attention mechanisms in sequential recommendations

Attention mechanisms have been successfully applied to various sequence modeling tasks, including machine translation [6], image captioning [21], and recommendation systems [7–10]. This allows the model to weigh the importance of different parts of the input sequence when making predictions. Many recent studies have embraced this architecture in the realm of sequential recommendation.

Kang et al. [7] took inspiration from this and presented a sequential model using self-attention called SASRec. Using self-attention, each item in the sequence serves as a query to calculate attention weight, effectively modeling dependencies between items in the sequence. Building on the achievements of the SASRec model, Alibaba Group [10] introduced BERT4Rec by making a bi-directional encoder model and applying it to sequential recommendation instead of using a left-to-right only in SASRec. BERT4Rec can learn the input sequence information from both sides and produce better performance. However, SASRec and BERT4Rec generally neglected timestamps, retaining only the order of items. Li et al. [9] argue that user interaction sequences should be modeled as a sequence with varying time intervals. Consequently, they introduced a novel TiSASRec model featuring a time-aware self-attention mechanism. This model considers not only the absolute positions of items like SASRec, but also the relative time intervals between any two items.

3 PROPOSED METHOD

In this section, we are going to demonstrate the process of building our proposed models based on SASRec and TiSASRec. We call the first proposed model EnrichedSASRec, which enriches the input representation of SASRec. Additionally, we designed a second model called EnrichedTiSASRec, which extends the time-aware attention mechanism of TiSASRec. The differences from the original models lie in the embedding and self-attention layers. Two novel embedding layers are designed to encode user ratings and item categories alongside the traditional item and timestamp sequences.

Figure 1 depicts the overall architecture of EnrichedTiSASRec. It contains three main layers: Embedding, Attention Blocks, and Prediction Layer. In addition to item and timestamp sequences, the model takes rating and category sequences. These sequences are encoded into representation vectors, which are then utilized in the attention layer to calculate the attention scores.

EnrichedSASRec has a similar architecture, but it does not have the time interval embedding, so it does not take the timestamps sequence as input.

3.1 Problem Formulation

Let U and I denote a set of users and items, respectively. In this research, explicit feedback is the user's item rating. For each user $u \in U$, the notation S^u represents the historical item sequence. We also employed the time modeling technique from [8] to model the time intervals among items in the time sequence T^u . In addition, we modeled a sequence of user ratings as F^u , a sequence of categories C^u . The input sequences are formalized as follows:

$$S^u = (s_1^u, s_2^u, \dots, s_{|S^u|}^u) \quad (1)$$

$$T^u = (t_1^u, t_2^u, \dots, t_{|T^u|}^u) \quad (2)$$

$$F^u = (f_1^u, f_2^u, \dots, f_{|F^u|}^u) \quad (3)$$

$$C^u = (c_1^u, c_2^u, \dots, c_{|C^u|}^u) \quad (4)$$

Where $s_i^u \in I$ is the interacted item at i , t_i^u is the timestamp when the user u interacts with the item i , f_i^u and c_i^u are the user rating and category of item i , respectively. During training, the goal of

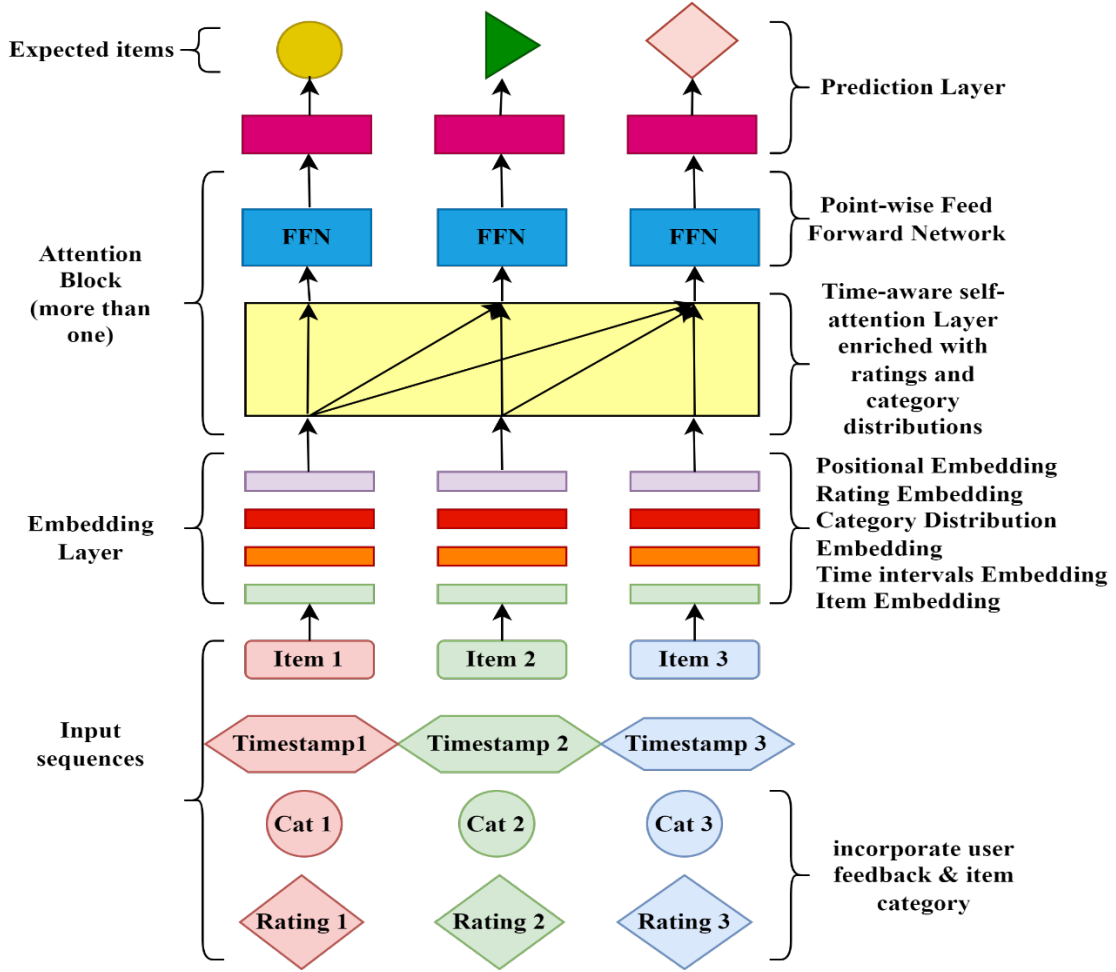


Figure 1: The EnrichedTiSASRec model architecture.

the model at time step t is to suggest the next item based on the last t items.

3.2 Embedding Layer

Following [7], the input sequence $(s_1^u, s_2^u, \dots, s_{|S^u|-1}^u)$ is transformed to a fixed-size sequence $(s_1, s_2, \dots, s_n)$ to ensure a consistent input size for the model regardless of the original length of the sequence. We have the item embedding matrix $M^I \in \mathbb{R}^{|I| \times d}$ which is used to encode the item sequence to $E^I \in \mathbb{R}^n \times d$, where $E_i^I = M_{s_i}^I$.

For EnrichedTiSASRec, the sequence of timestamps is processed like [6] to retrieve a time sequence $(t_1, t_2, \dots, t_n)$ and then it is used to calculate the time relation matrix. Two embedding matrices $M_K^R \in \mathbb{R}^{k \times d}$, $M_V^R \in \mathbb{R}^{k \times d}$ are employed to encode the time relation to $E_K^R \in \mathbb{R}^n \times n \times d$, $E_V^R \in \mathbb{R}^n \times n \times d$, where k is the maximum relative time interval between two items. We also use two separate positional embedding matrices $M_K^P \in \mathbb{R}^n \times d$, $M_V^P \in \mathbb{R}^n \times d$. After retrieval, we have $E_K^P \in \mathbb{R}^n \times d$, $E_V^P \in \mathbb{R}^n \times d$.

Our proposed information about user feedback, and category distribution are also processed similarly, resulting in a rating sequence $(f_1, f_2, \dots, f_n)$, and a sequence of categories $(c_1, c_2, \dots, c_n)$. We employ the rating embedding matrix $M^F \in \mathbb{R}^{f \times d}$ and the category embedding matrix $M^C \in \mathbb{R}^{z \times d}$, where f is the maximum score of rating, which scales from 1 to 5, z is the maximum number of categories in the dataset, and d is the latent dimensionality. The rating and category sequences are encoded using the embedding look-up operation to retrieve four embedding matrices $E_K^F, E_V^F, E_K^C, E_V^C \in \mathbb{R}^n \times d$, where $E_i^F = M_{s_i}^F$, $E_i^C = M_{c_i}^C$. Note that for each type of information, we also use two separate embedding matrices for keys and values in the attention mechanism. The input representation contains item embedding, time interval embedding, positional embedding and is enriched by rating and category embeddings. These embeddings are used to calculate the attention score in the attention mechanism.

With the EnrichedSASRec, the difference in this layer is that it does not take the time sequence $(t_1, t_2, \dots, t_n)$ as input, so it ignores everything related to the time relation among items.

3.3 Feature-enriched Attention

Drawing from the innovative concept of relative position [7] and time interval-aware self-attention mechanisms [8], our study introduces a novel extension of the attention mechanism by incorporating the rating score and categorical information of each item.

3.3.1 Feature-enriched Attention Layer. For each input sequence $E^I = (m_{s_1}, m_{s_2}, \dots, m_{s_n})$ of n items $m_{s_i} \in \mathbb{R}^d$, the output sequence Z is computed: $Z = (z_1, z_2, \dots, z_n)$, where $z_i \in \mathbb{R}^d$. The representation for each element z_i in the output sequence is calculated as a weighted sum of the linearly transformed embeddings of the input sequence elements, along with the embeddings capturing the time interval relation, position, ratings, and categories:

$$z_i = \sum_{j=1}^n \alpha_{ij} (m_{s_j} W^V + r_{ij}^v + p_j^v + f_j^v + c_j^v) \quad (5)$$

Where $W^V \in \mathbb{R}^{d \times d}$ is the input projection for value; $m_{s_j}, r_{ij}^v, p_j^v, f_j^v, c_j^v$ are the item embedding, time relation between item j and item i , position, rating, and category of item j , respectively.

The weight coefficient α_{ij} is computed using the softmax function:

$$\alpha_{ij} = \text{softmax} \left(\frac{m_{s_i} W^Q (m_{s_j} W^K + r_{ij}^k + p_j^k + f_j^k + c_j^k)^T}{\sqrt{d}} \right) \quad (6)$$

Where $W^Q \in \mathbb{R}^{d \times d}$, $W^K \in \mathbb{R}^{d \times d}$ is the input projection for query and key, respectively. The attention score is calculated using the scaled-dot product attention similar to the original transformer model [6] where $Q_i = m_{s_i} W^Q$, $K_i = m_{s_j} W^K + r_{ij}^k + p_j^k + f_j^k + c_j^k$, and $V_i = m_{s_j} W^V + r_{ij}^v + p_j^v + f_j^v + c_j^v$.

Note that the formulas (5) and (6) are for the EnrichedTiSAS-Rec model. Since the EnrichedSASRec does not consider the time interval, the formulas became:

$$z_i = \sum_{j=1}^n \alpha_{ij} (m_{s_j} W^V + p_j^v + f_j^v + c_j^v) \quad (7)$$

$$\alpha_{ij} = \text{softmax} \left(\frac{m_{s_i} W^Q (m_{s_j} W^K + p_j^k + f_j^k + c_j^k)^T}{\sqrt{d}} \right) \quad (8)$$

The scale factor $\sqrt{d}$ is employed to avoid large values in the inner product, which is particularly crucial in scenarios where the dimensionality is high. Like the approach in [7], we employ a masking strategy to ensure that the representation at each time step is only influenced by previous items in the sequence, adhering to the sequential nature of the recommendation task.

3.3.2 Point-Wise Feed-Forward Network. To incorporate nonlinearity and address interactions among various latent dimensions, a point-wise feed-forward network is uniformly applied to the output Z of the attention layer:

$$F_i = \text{FFN}(z_i = \text{ReLU}(z_i W^{(1)} + b^{(1)})) W^{(2)} + b^{(2)} \quad (9)$$

Where $W^{(1)}, W^{(2)}$ are $d \times d$ matrices, $b^{(1)}, b^{(2)}$ are d -dimensional vectors.

As the network becomes complicated, overfitting and vanishing gradients occur. Dropout is employed:

$$Z_i = z_i + \text{Dropout}(\text{FFN}(\text{LayerNorm}(z_i))) \quad (10)$$

Layer normalization is also applied to stabilize training, accelerate convergence, and reduce sensitivity to initial weight configurations:

$$\text{LayerNorm}(x) = \alpha \odot \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta \quad (11)$$

Where $\odot$ is an element-wise product, μ and σ are the mean and variance of x , α and β are the learned scaling factors and bias terms, respectively.

An *attention block* is constructed by a combination of a feature-enriched attention layer with a point-wise feed-forward network. We stack more than one block to capture more complicated patterns.

3.4 Prediction Layer

Following the stacking of self-attention blocks, the model generates a combined representation that encapsulates information about items, their positions, time differences among items, user ratings, and categories. The relevance score of the next item i is computed by adopting a Matrix Factorization method to predict the next item. Specifically, the representation of the previous t items Z_t is taken inner product with the embedding M_i^I of item i as follows:

$$R_{i,t} = Z_t M_i^I \quad (12)$$

The relevance score $R_{i,t}$ of each candidate item i is the probability of it being the next item after the first t items. With this score, we can rank the items, determine the top- k items, and recommend them to the users.

3.5 Model Inference

The model input is four sequences: an item sequence $S^u = (s_1^u, s_2^u, \dots, s_{|S^u|-1}^u)$, a time sequence $T^u = (t_1^u, t_2^u, \dots, t_{|T^u|-1}^u)$, a rating sequence $F^u = (f_1^u, f_2^u, \dots, f_{|F^u|-1}^u)$, and a category sequence $C^u = (c_1^u, c_2^u, \dots, c_{|C^u|-1}^u)$. They become $(s_1, s_2, \dots, s_n)$, $(t_1, t_2, \dots, t_n)$, $(f_1, f_2, \dots, f_n)$, and $(c_1, c_2, \dots, c_n)$, respectively, via truncation or padding. At time step t , we define the expected output o_t :

$$o_t = \begin{cases} \langle \text{pad} \rangle, & \text{if } s_t \text{ is a padding item } (\langle \text{pad} \rangle \text{ indicates a padding item}) \\ s_{t+1}, & \text{if } 1 \leq t < n \\ s_{|S^u|}^u, & \text{if } t = n \end{cases} \quad (13)$$

The output is a shifted sequence $(s_2^u, s_3^u, \dots, s_{|S^u|}^u)$. Like [7], for each input sequence, a negative item o'_t is paired with the expected positive item o_t to optimize the loss function, which employs the Binary Cross-Entropy Loss:

$$-\sum_{S^u \in S} \sum_{t \in \{1, 2, \dots, n\}} [\log(\sigma(r_{o_t, t})) + \log(1 - \sigma(r_{o'_t, t}))] + \lambda \|\Theta\|_F^2 \quad (14)$$

Where r_{o_t} and $r_{o'_t}$ are the preference scores of positive item o_t and negative item o'_t , respectively, that are transformed into the range $(0, 1)$ using the sigmoid function σ . $\Theta = \{M^I, M_K^P, M_V^P, M_K^R, M_V^R, M_K^F, M_V^F, M_K^C, M_V^C\}$ is the set of embedding matrices respectively to items (I), positions (P), time relations

(R), ratings (F), categories (C), $||\cdot||_F$ represents the Frobenius norm, and λ is the regularization parameter.

The model is optimized using the Adam optimizer [22], a variant of Stochastic Gradient Descent (SGD) that incorporates adaptive moment estimation.

4 EXPERIMENTS

This section outlines the experimental methodology and presents the empirical evaluation results. Our experiments aim to investigate the following research questions:

- **RQ1:** How does the recommendation performance of our proposed method compare against the original models SASRec and TiSASRec for sequential recommendation tasks?
- **RQ2:** Does the incorporation of user ratings and item category information provide tangible benefits in enhancing the recommendation accuracy and personalization capabilities of the model?

4.1 Dataset

This study uses real-world data from the Amazon Review dataset. It comprises 233.1 million reviews from May 1996 and Oct 2018. We focus on the 5-core version to get the user-item interactions. The dataset also includes extra information like reviews and product metadata. In this experiment, we take the rating file and then join them with the metadata file to determine the category of each item. Each use-item interaction includes the information in the following order:

- *itemID*: The ID of the rated product, which is an ASIN that stands for Amazon Standard Identification Number.
- *userID*: The ID of the user who rated the product.
- *rating*: The rating score of a user on an item on the five scales.
- *timestamp*: The time that the user rates the product. This helps us get the user sequence in chronological order.

For preprocessing the data, we follow the same procedures and techniques employed in the prior works [7, 8]. The initial step involves eliminating interaction with ratings below three. The next step is to eliminate users and items with less than thirty actions in the datasets. The purpose of this is to remove outliers within the dataset and ensure that the length of the interaction sequence for each user is sufficient for the model to identify the relationships among the interactions.

To feed data into the model, particularly when dealing with categorical variables such as *userID*, *itemID*, and *category*, it is essential to convert these entities into indexes that the model can interpret. We created three lookup tables for users, items, and categories. Then, we replace each *userID*, *itemID*, and *category* in the dataset with their corresponding indexes using those tables.

After the preprocessing, the total number of interactions is reduced from more than 50 million to 4,412,181, but the amount of data is still significant for assessing the recommendation performance. The statistics of the dataset after preprocessing are presented in Table 1. Note that the Amazon dataset is already quite clean, with no missing values.

Table 1: The statistics of the Amazon Books dataset

Statistics	Values
Users	57,208
Items	51,022
Interactions	4,412,181
Avg. Sequence Length	75.13
Sparsity	99.84%
Avg. Interaction/Item	86.47

4.2 Evaluation Metrics

To measure the model’s effectiveness, we adopt the same metrics from the original papers, which are HR@K and NCDG@K [23] [24] [25]. The value of k is set to 10 and 20. The Hit Rate metric (HR@K) measures the accuracy of a model in including the correct items in the recommended K items. Normalized Discounted Cumulative Gain (NDCG@K) evaluates the suggestion by considering the items in the top-K recommendation list’s rank order and relevance.

4.3 Implementation Details

In our approach, we prepare the dataset by partitioning it into training, validation, and test sets. Each sequence is segmented such that the last item serves as the test set, the second last as the validation set, and the remaining items for training. To enhance training efficiency and mitigate data sparsity challenges, we employ a negative sampling technique used in [7].

Hyperparameter optimization is crucial to our model tuning strategy. We focus on two critical hyperparameters: *hidden_units*, representing the latent dimensionality in the embedding layer, and *num_blocks*, indicating the number of attention blocks. A grid search is conducted over *hidden_units* = [32, 64, 128] and *num_blocks* = [2, 3], evaluating each combination against the NDCG@10 metric over 50 epochs. Following this optimization, we determine that *hidden_units*=64 and *num_blocks*=2 yield optimal performance and proceed with these settings for further training.

During training, all models are configured with a maximum sequence length of $n=100$ to capture relevant user interactions effectively. Given the inherent sparsity of the dataset, *dropout_rate* = 0.4 is applied to enhance generalization and prevent overfitting. Due to computational constraints and the complexity of model training, we conduct training sessions for 100 epochs with *batch_size* = 128. The Adam optimizer, configured with a learning rate of 0.001, ensures consistency across experiments and facilitates efficient convergence.

Our implementation uses the PyTorch framework, utilizing an NVIDIA Tesla T4 GPU with 16GB of memory.

4.4 Compared Methods

Since this study focuses on how the effectiveness is improved after modifying SASRec and TiSASRec, other relevant methods are omitted in this experimental comparison.

The experimental results are shown in Table 2, the top-performing method for each metric is highlighted in bold, while the second-best method is underlined. (**RQ1**).

Table 2: The performance comparison of models

	NDCG@10	HR@10	NDCG@20	HR@20
SASRec	0.5728	0.8226	0.5956	0.9122
EnrichedSASRec	0.5610	0.8117	0.5851	0.9062
TiSASRec	<u>0.6633</u>	0.8641	<u>0.6790</u>	0.9257
EnrichedTiSASRec	0.6663	<u>0.8635</u>	0.6821	<u>0.9255</u>

To find out the effectiveness of the novel rating and category embeddings, we compared the performance of our proposed models with their original models:

- SASRec and EnrichedSASRec: The addition of rating and category embeddings in EnrichedSASRec leads to a decrease in for metrics NDCG @10 (from 0.5728 to 0.5610), NDCG @20 (from 0.5956 to 0.5851), HR@10 (from 0.8226 to 0.8117) and HR@20 (from 0.9122 to 0.9062) compared to the original SASRec model.
- TiSASRec and EnrichedTiSASRec: The incorporation of rating and category embeddings in EnrichedTiSASRec leads to an improvement in NDCG@10 (from 0.6633 to 0.6663) and NDCG@20 (from 0.6790 to 0.6821). However, it does not perform better than TiSASRec on HR@10 (from 0.8641 to 0.8635) and HR@20 (from 0.9257 to 0.9255).

The results suggest that the incorporation of rating and category embeddings does not significantly impact the performance of the original SASRec model. However, these embeddings incorporated with the time interval embedding in TiSASRec have the potential to enrich the representation of the input data, which could lead to improved recommendation performance in the long run (RQ2).

EnrichedTiSASRec demonstrates a well-rounded performance by leveraging the synergy between rating embeddings and category embeddings. The rating embeddings effectively capture user preferences, while the category embeddings provide additional contextual information about the items. This combination of complementary information sources enables the model to develop a comprehensive understanding of user preferences and item relationships, leading to more accurate recommendations on the ranking metric. Although the result on hit rate metric is a bit lower than TiSASRec, EnrichedTiSASRec maintains competitive performance in this aspect.

5 CONCLUSIONS

This research investigated the effectiveness of incorporating rating and category embeddings into the SASRec and TiSASRec models. The experimental results demonstrated that the model combining rating and category embeddings, EnrichedTiSASRec, performed better than the other models in terms of ranking while maintaining competitive hit rate quality. The rating embeddings played a crucial role in capturing user preferences, leading to improved recommendation accuracy. Additionally, the category embeddings provided a complementary source of information, further enhancing the retrieval of relevant items. While these findings are promising, future work could explore the impact of different embedding techniques and model architectures on recommendation performance. Furthermore, incorporating additional contextual information, such as

temporal dynamics or social connections, may lead to even more personalized and accurate recommendations.

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